



M.SC. DATA ANALYTICS & TECHNOLOGIES

DAT7301- DATA ANALYSIS AND VISUALISATION

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Abstract

This study examined the relationship between seafood production, trade volumes, and production patterns in the UK using an observational study design. The research explored key questions regarding the dominant seafood species in imports and exports, the correlation between production and trade, and future projections for UK seafood production. A comprehensive dataset covering 2009–2020 was utilised, incorporating data from sources such as the Marine Management Organisation (MMO), HM Revenue and Customs (HMRC) and the Centre for Environment, Fisheries and Aquaculture Science (CEFAS). Statistical methods including descriptive analysis, correlation tests and time-series forecasting were employed to assess the trends in seafood trade. Potential sources of errors and biases were identified, including inconsistencies in data reporting, misclassification of foreign fleets and selection bias. Ethical considerations such as data privacy, integrity and transparency were addressed to ensure compliance with research standards. The findings highlight the need for improved data harmonisation, strategic policy interventions and sustainable trade practices to enhance seafood supply chain. The study insights contribute to evidence-based decision-making for policymakers, businesses and stakeholders in the UK seafood industry.

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Chapter One

1.1 Introduction

Data analysis and visualisation are essential components of modern decision-making across various industries. These techniques help transform raw data into meaningful insights, allowing businesses and researchers to make informed decisions. To gain insights, specialised tools and techniques must be used. The structured analysis of data to identify patterns and draw conclusions is referred to as data analytics (Hyman *et al.*, 2024).

The data analytics process differs depending on the situation and the tools available, but it usually follows a clear, step-by-step process that starts from defining the problem and the main questions to answer, to collecting and cleaning the data, performing statistical analysis and finally presenting the data with visuals (Hyman *et al.*, 2024). Data visualisation is a mix of creativity and technical skills, where you need to find the right balance between making something look good and making sure it is accurate. The most important job of any chart or graph is to show the data as it truly is, without any misleading or skewed representations, an effective visualisation enhances its message rather than detracting from it (Wilke, 2019).

Using data to drive decisions ensures that choices are based on real insights. By spotting trends, anomalies and patterns, data analysis helps reduce mistakes and leads to more reliable conclusions. Data-driven problem-solving make it easier to focus resources like time and budget where they would have the maximum impact. By leveraging data analytics, organisations can optimise processes, improve performance, develop strategic initiatives, aid hypothesis testing, forecasting, and evaluating outcomes, making it a powerful tool for addressing real-world challenges (Camm *et al.*, 2023).

Chapter Two

2.1 Research Problem

Understanding how domestic production relates to trade can inform government policies on food security, trade agreements, and sustainability in the fishing industry. The research question focuses on the relationship between production and trade volume. It is crucial for policymakers, businesses, and stakeholders in the seafood industry to optimise supply chains, forecast demand, and assess policy impacts.

- i. Which species dominate the UK seafood exports and imports
- ii. What is the correlation between seafood production and total trade volume
- iii. What would be the UK seafood production in five years

Chapter Three

3.1 Study Design

The study will use an observational study design which studies collected or past data to understand patterns and use statistical methods and machine learning techniques to identify relationships between variables (Ranganathan, 2019). Steps in an observational study include:

- I. Define the Research Question: Identify specific, measurable research objectives
- II. Select the Population and Sampling Method: Choose appropriate data sources and sampling techniques
- III. Collect and Clean Data: Ensure accurate and complete data collection from verified sources
- IV. Identify Confounding Variables: Determine external factors that may influence relationships
- V. Perform Statistical Analysis: Apply suitable techniques like regression modelling, time-series analysis, and hypothesis testing
- VI. Interpret and Validate Findings: Compare results with previous studies to ensure reliability

3.2 Target population and sampling method

The target population for this analysis is the all the seafood trade data in the UK from 2009 to 2020.

The sampling method is a census method as the dataset covers a comprehensive UK trade data and all the entries would be used for analysis (Groves, 2009).

3.3 Sample size and sample power

The dataset consists of 2,691 records and would be sufficient for trend and comparative analysis, the sample size is 2,691 and the sample power is 1.0 or 100%

3.4 Variables and measurement method

Independent variables: Calendar year, commodity type, species, species type

Dependent variables: Trade volume measured in grams

3.5 Data Collection Approach

According to (Löfstedt *et al.*, 2025) “We created a unique database linking seafood production (capture and aquaculture), trade (imports and exports), purchases (both within and out-of-home), and UK consumption data, to species level, with data obtained between 2009 and 2020 (the most recent year all datasets were available to).” The data was sourced from

- I. Marine Management Organization (MMO) landings data were used to calculate the total volumes of each species landed into the UK and abroad by UK vessels.
- II. Total volumes of farmed seafood produced in UK waters were obtained from the Centre for Environment, Fisheries and Aquaculture Science (CEFAS).
- III. Annual import and export volume of all seafood and products were obtained from HM Revenue and Customs (HMRC) trade data.
- IV. Average weekly per capita UK seafood purchases were obtained from the Department for Environment, Food and Rural Affairs (DEFRA) family food data.

Chapter Four

4.1 Sources of Errors and Biases

Every data has potential for error or uncertainty in the data collection process, the UK supply chain dataset has some potential for error which include:

- I. Differences in conversion factors used by HMRC and MMO for the same species could introduce discrepancies in reported seafood volumes.
- II. Exact volumes of seafood stored in cold storage and volumes wasted at different supply chain levels are unknown, leading to potential misestimation of available seafood supplies.
- III. There are inconsistencies in definitions of foreign fleets between HMRC and MMO, which may lead to misclassification errors in trade data.
- IV. The study relies on different datasets from organisations such as the Marine Management Organisation (MMO), HM Revenue and Customs (HMRC), the Centre for Environment, Fisheries and Aquaculture Science (CEFAS), and other relevant sources.

4.2 Common biases from the study design or data

When analysing UK supply chain data, various biases may arise due to the study design, data sources, and external influences. Below are some of the most common biases and strategies to mitigate them:

- | | |
|--|------|
| I. Selection | Bias |
| <p>Selection bias occurs when the sample data is not representative of the overall population (Groves, 2009). This can happen if the dataset primarily includes data from large corporations while excluding small and medium-sized enterprises as the study focuses on major commercially important species, potentially overlooking other less-consumed fish. The results may overestimate</p> | |

or underestimate trends in supply chain efficiency and conclusions may not be generalizable to all supply chain stakeholders

II. Response Bias

Response bias occurs when survey or self-reported data is systematically inaccurate due to misreporting, social desirability, or recall errors. Firms may exaggerate supply chain resilience or downplay disruptions and the data might reflect perceptions rather than actual performance (Philip M. Podsakoff, Scott B. MacKenzie and Nathan P. Podsakoff, 2012).

III. Confounding variables

Confounding variables occurs when an external factor influences both the dependent and independent variables. An analysis could mistakenly link supply chain inefficiencies to Brexit, when broader global shipping disruptions may have a more significant impact. Similarly, the effect of inflation on logistics costs might be obscured by other underlying economic influences (Rosenbaum, 2017).

4.3 Strategies to minimize common biases

Mitigation Strategies include:

- I. Use random sampling techniques where possible to ensure a diverse representation of firms.
- II. Expand Data Sources: Supplement existing datasets with information from industry reports, trade associations, and government agencies to include a wider range of species and supply chain participants.
- III. Cross-check self-reported data with objective third-party data and government databases to ensure accuracy
- IV. Design anonymous surveys to encourage honest responses.

- V. Encourage firms to follow consistent and transparent reporting standards to minimise discrepancies in data submission
- VI. Compare trade patterns before and after key global to differentiate between their individual effects.

Chapter Five

5.1 Step by Step Statistical analysis plan

- I. Data Cleaning and Preparation: Remove missing values and standardise data formats, convert volume measurements to a consistent unit (kilograms) and aggregate trade volume by year, species, and trade type (Han, Pei and Tong, 2023).
- II. Descriptive Statistics: Calculate mean, median, and standard deviation of trade volume for imports, exports, and production. Generate frequency distributions for different seafood species and visualise trends using histograms and boxplots (Field, 2018).
- III. Inferential Statistics: Use correlation analysis to examine the relationship between seafood production and total trade volume. Hypothesis testing to determine if trade volumes differ significantly between different species (Montgomery and Runger, 2018). Conduct t-tests to compare trade volumes before and after major events.
- IV. Advanced Methods: Time-Series Analysis to conduct trend analysis and evaluate seafood trade patterns over time. Apply ARIMA models to forecast future trade volumes (Chatfield and Xing, 2019).

5.2 Justification for Statistical Methods

- I. Descriptive statistics provide insights into the overall distribution and patterns within the dataset (Field, 2018). By computing and visualising these statistics, researchers can detect anomalies, identify trends, and ensure data integrity before conducting more advanced analyses.
- II. Correlation analysis is appropriate for assessing relationships between seafood production and trade volume (Moore, McCabe and Craig, 2021).

- III. Hypothesis testing determines whether observed differences in trade patterns are statistically significant (Montgomery and Runger, 2018).
- IV. Time-series analysis is crucial for understanding historical trends and making future projections (Chatfield and Xing, 2019).

Chapter Six

6.1 Ethical concerns

Ethical concerns in our dataset include:

Data Privacy

The study utilises data from multiple sources including household food purchase records. Although aggregated, these datasets may contain sensitive personal information related to dietary habits and consumer behaviour. Unauthorized access to raw datasets and misuse of consumer information beyond the intended research purposes can present potential risks (Löfstedt *et al.*, 2025).

Data Integrity

The study compiles data from multiple agencies (MMO, HMRC, DEFRA, CEFAS), each with different definitions, measurement techniques, and reporting standards. Ethical concerns arise if inconsistent or incomplete datasets are used which would lead to misinterpretation of seafood supply trends and potentially misleading policy recommendations (Löfstedt *et al.*, 2025).

Transparency

Differences in classification systems and conversion factors like how different agencies define foreign fleets or estimate edible portions of seafood can lead to data misrepresentation. Ethical concerns include the risk of bias in data processing and lack of transparency in how missing data were handled (Löfstedt *et al.*, 2025).

6.2 Measures to ensure ethical compliance

- I. Use only aggregated and anonymised datasets to ensure individual privacy is maintained (Solove, 2004).

- II. Comply with data protection regulations such as the UK General Data Protection Regulation (UK GDPR) when handling consumer related data (Kuner *et al.*, 2020).
- III. Restrict access to raw data and ensure it is stored securely with encryption and controlled access permissions (Yu *et al.*, 2024).
- IV. Clearly document data sources, methodologies, and limitations to maintain transparency in data interpretation (Open Data Institute, 2019).
- V. Conduct peer reviews and independent audits of the data aggregation process to verify accuracy and mitigate bias (Rosenbaum, 2017).

Chapter Seven

7.0 Conclusion

This study examined UK seafood trade and production trends, focusing on the dominant species in imports and exports, the correlation between production and trade volume, and future seafood production projections. The research utilised an observational study design, analysing historical data from 2009 to 2020 sourced from MMO, HMRC, CEFAS, and DEFRA. The study followed a structured data analysis process, incorporating data cleaning, descriptive statistics, correlation analysis, hypothesis testing, and time-series forecasting to extract meaningful insights.

The research recognised potential sources of error and biases, such as differences in conversion factors, missing data on cold storage and food waste, inconsistencies in trade classifications, and data entry mistakes. Additionally, selection bias, response bias, and confounding variables were acknowledged as possible influences on the accuracy of the study.

To mitigate these issues, strategies such as random sampling, expanding data sources and cross-checking self-reported data were proposed. The statistical analysis plan was designed to ensure data integrity and meaningful interpretation, incorporating descriptive statistics, correlation analysis and time-series modelling to assess seafood trade patterns.

The study also addressed ethical considerations, ensuring data privacy, compliance with UK GDPR regulations, transparency in data aggregation, and accuracy in reporting. Measures such as data anonymisation, restricted access, peer reviews, and independent audits were recommended to maintain ethical research standards.

In conclusion, this study highlights the importance of structured data analysis and statistical methods in understanding seafood trade dynamics. The use of observational study design and quantitative techniques

provides valuable insights for policymakers and industry stakeholders, allowing for more informed decision-making regarding seafood supply chain management and trade policies.

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